

Multilingual Speech-to-Text Subtitle Generator using Whisper Large-V3

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Abstract:

The manual creation of subtitles for audio and video content is a labor-intensive, time-consuming, and error-prone process, creating a significant barrier to accessibility and content creator productivity.

This paper presents an end-to-end automated pipeline for generating accurate, synchronized subtitle files (.srt) from diverse media inputs. Our system integrates a multi-stage preprocessing module for audio enhancement, including noise reduction, voice activity detection, and bandpass filtering, to condition real-world audio for optimal transcription.

The core transcription is handled by an optimized implementation of OpenAI's Whisper model (faster-whisper large-v3), chosen for its state-of-the-art multilingual accuracy and robustness. A dedicated post-processing module refines the transcribed text by removing disfluencies and normalizing Indic scripts.

Experimental results on English, Hindi, and French content under various acoustic conditions demonstrate the system's high accuracy and synchronization capabilities. The proposed solution provides a cost-effective, scalable, and efficient alternative to manual subtitling and existing commercial tools, significantly enhancing media accessibility.

Keywords—Automatic Speech Recognition (ASR), Subtitle Generation, Whisper, Natural Language Processing, Audio Preprocessing, Accessibility

Introduction:

The proliferation of video content across platforms like YouTube, educational portals, and streaming services has created an unprecedented demand for subtitles. Subtitles are crucial not only for the deaf and hard-of-hearing community [1] but also for language learners, viewers in sound-sensitive environments, and for improving overall content comprehension. Traditionally, subtitle creation is a manual process involving listening, transcribing, and painstakingly synchronizing text with audio—a process that can take 5 to 10 times the duration of the video itself.

While automated solutions exist, they often suffer from critical limitations. Platform-native tools like YouTube Auto-caption have limited language support and struggle with accents or technical jargon. Commercial services like Rev and [Otter.ai](#) offer high accuracy but are cost-prohibitive for individual creators or small-scale applications. Many open-source alternatives lack the robustness to handle noisy audio or provide poor synchronization.

This paper addresses these challenges by presenting a robust, open-source Automatic Subtitle Generator. The primary contributions of this work are:

1. The design and implementation of a comprehensive audio preprocessing pipeline that significantly enhances input quality for challenging, real-world audio conditions.
2. The integration of an optimized, large-scale multilingual ASR model (Whisper large-v3) within a chunking framework to handle long-duration files efficiently.
3. The development of language-aware text post-processing techniques, including specialized normalization for Indic scripts.
4. An end-to-end system that accepts a wide range of audio/video formats and produces broadcast-ready subrip (.srt) subtitle files.

Prior Related Work:

Several ASR systems have been developed over the past decade:

Google Speech-to-Text

Provides strong English transcription but struggles with Indian accents, background noise, and mixed languages.

Mozilla DeepSpeech

A pioneering open-source ASR model, but its accuracy drops for long-form speech and multilingual content.

Facebook wav2vec 2.0

A self-supervised model that significantly improved ASR accuracy but requires fine-tuning for Indian languages.

YouTube's Subtitle System

High-quality but closed-source. Limited support for regional Indian languages for offline or custom use.

OpenAI Whisper

Whisper introduced a new paradigm with multitask training (transcription, translation, segmentation).

Its **Large-V3** model provides:

- High robustness to noise
- Over 100 languages
- Stable timestamps
- Multilingual accuracy

However, raw Whisper text often contains filler words, punctuation issues, or Unicode errors—especially in Indic scripts.

Our project specifically improves Whisper's output using post-processing methods.

Dataset:

This project does **not rely on a pre-defined static dataset**.

Instead, we use a **dynamic dataset** consisting of user-uploaded audio/video files.

Sources of Data:

- Recorded lectures
- Public speeches
- Conversations
- YouTube clips
- Real-world recordings from phones
- Multilingual content (Telugu, Hindi, English, Tamil, etc.)

Characteristics:

- Noisy environments
- Variable microphone quality
- Mixed code languages
- Long continuous recordings

Example Stats from a Sample Audio:

- Original duration: **433.58 seconds**
- After VAD (silence removal): **287.2 seconds**
- Speech segments detected: **518**
- Number of chunks (30s each): **10**
- Final SRT entries: **177**

This dataset style ensures the system generalizes well to real-life audio conditions.

Methodology / Model:

A. Preprocessing Pipeline

1. File Conversion (FFmpeg)

All audio/video formats are standardized to **16 kHz mono WAV**, ensuring reliable input for Whisper.

2. Noise Reduction

We use the *noisereduce* library with `prop_decrease = 0.9` to aggressively suppress background noise.

3. Silence Removal (VAD)

Using `librosa.effects.split`, non-speech intervals are discarded to reduce model hallucinations and processing time.

4. Bandpass Filtering

Human speech predominantly lies between 70 Hz–8000 Hz. A 4th-order Butterworth bandpass filter eliminates out-of-range noise.

5. Normalization

Audio amplitude is normalized to prevent clipping and ensure consistent loudness.

6. Chunking

The cleaned audio is split into 30-second chunks to avoid GPU memory limits and improve transcription accuracy.

B. Speech-to-Text (Whisper Large-V3)

Whisper is an encoder-decoder Transformer trained on a massive multilingual dataset. For each chunk, Whisper outputs:

- Detected language
- Timestamped transcriptions
- Probability scores

Whisper Large-V3 was chosen due to:

- High multilingual accuracy
- Strong robustness to noise

- Ability to handle accents and code-switching
- GPU support for fast inference

C. Text Post-Processing

1. General Cleaning

We remove:

- Filler words (e.g., *um*, *uh*)
- Repeated punctuation
- Extra whitespace
- Special characters

2. Indic Script Normalization

Languages such as Hindi, Telugu, and Tamil use virama-based characters.

We fix issues like:

ॐ ँ → ॐ

Zero-width joiners and unicode artifacts are also removed.

D. SRT Subtitle Generation

Each segment is converted into the SRT format:

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00:00:03,120 --> 00:00:05,540

Hello everyone, welcome.

The timestamp offset from chunking is correctly added to maintain global timing.

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Experiments:

Datasets and Testing Methodology: The system was evaluated on a diverse set of real-world audio samples to test its robustness.

- **English:** Clean audio from TED Talks.
- **Hindi:** Casual speech from the "Suhani Shah" show, featuring natural accents and conversational language.
- **French:** Challenging, noisy field recordings.

Evaluation Metrics: Performance was assessed based on:

- **Transcription Accuracy:** Approximated using qualitative analysis and spot-checking against manual transcripts.
- **Timestamp Synchronization:** Visually verified in video players.
- **Processing Speed:** Measured end-to-end on a Google Colab GPU instance.
- **Format Robustness:** Tested with all declared input formats.

Results:

The system successfully processed all input formats and generated well-synchronized SRT files. The Whisper large-v3 model demonstrated high accuracy, meeting the target of approximately 90% correct transcription on clean and moderately noisy audio. A key result was the model's automatic and accurate language detection, as evidenced by a high probability score (>0.98) for French audio in the provided code output.

The preprocessing pipeline proved highly effective. For instance, on a 433-second input file, VAD successfully identified 518 speech segments and reduced the audio to be processed to 287 seconds, a 34% reduction in processing time and potential noise.

Qualitative System Performance Across Languages

LANGUAGE	AUDIO CONDITION	TRANSCRIPTION ACCURACY	SYNCHRONIZATION	KEY CHALLENGES
English	Clean (TED Talk)	Very High (~95%)	Excellent	None
Hindi	Casual, Accented	High (~85%)	Good	Indic Script Normalization
French	Noisy (Field)	Moderate-High (~80%)	Good	Background Noise
Italian	Song	Moderate(~70%)	Moderate	Tunes and Background music
Telugu (mix)	Song	Low(~50%)	Low	Same wording in different languages.

Analysis & Conclusion:

A. Strengths

The project's primary strength lies in its integrated pipeline that combines robust preprocessing with a state-of-the-art ASR model. This synergy allows it to handle real-world audio that would challenge standalone Whisper implementations. The use of faster-whisper makes the system practical by providing high accuracy without prohibitive computational cost. Furthermore, the system is truly multilingual and offers a cost-effective solution compared to commercial APIs.

B. Limitations and Future Work

Despite its successes, the system has limitations. Processing very long, high-quality audio files can strain the memory constraints of free-tier cloud environments like Google Colab. Accuracy can still degrade with extremely heavy, non-stationary background noise (e.g., loud music or overlapping conversations). While robust, subtle regional accents can occasionally lead to transcription errors.

Future work will focus on:

1. Implementing a dynamic chunking strategy to better handle memory limits.
2. Integrating a more powerful, non-stationary noise reduction algorithm.
3. Developing a user-friendly Graphical User Interface (GUI) for desktop use.
4. Exploring fine-tuning the Whisper model on a domain-specific dataset to further improve accuracy for specialized vocabulary.

Conclusion

This paper presented a complete, functional pipeline for the automatic generation of subtitles from audio and video files. By leveraging a sophisticated audio preprocessing stage and the powerful, optimized Whisper large-v3 model, the system demonstrates high accuracy and robustness across multiple languages and challenging audio conditions. The end product is a practical, cost-effective tool that significantly reduces the burden of manual subtitle creation, thereby enhancing the accessibility and reach

of multimedia content. The project successfully bridges the gap between cutting-edge ASR research and a tangible, user-facing application.

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